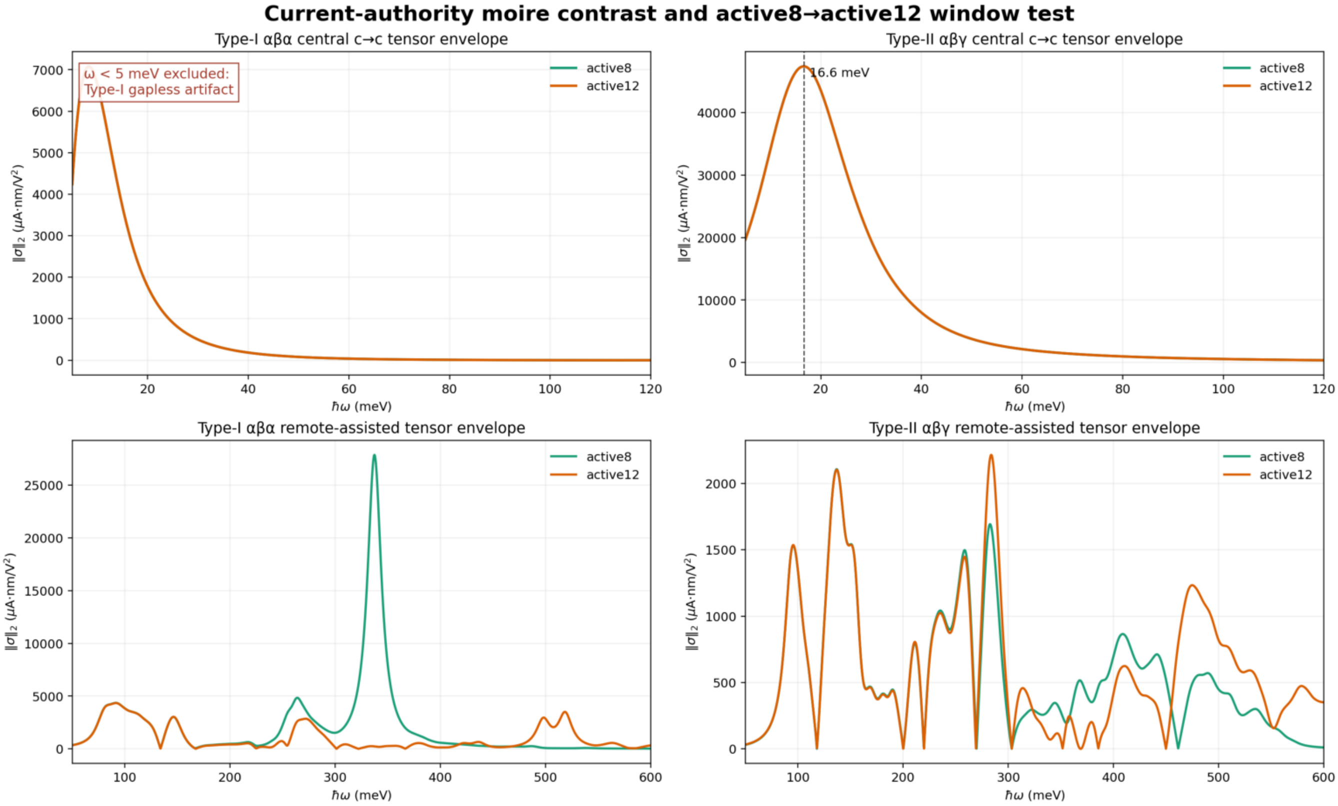
HTQG 巨大体光伏效应

拓扑、序参量与工程量级

Revision: zero-frequency exclusion + thickness conversion



• Type-I αβα 的错误零频大电流已从所有 revised frequency plots 排除（hbarω < 5 meV）。

• Type-II αβγ clean finite-frequency response 在 16.6-17.8 meV，且通过 mesh/window checks。

• Largest scanned peak = 6.876e4 μA·nm/V²；P2-approved headline = 2.3678e4 μA·nm/V²。

• 按 0.67-1.0 nm 厚度估算，approved component 等效为 2.37e4-3.53e4 μA/V²。

2026-08-04 | figure-led report | no raw data included

先排除错误零频，再比较真实 finite-frequency BPVE

- Neutral Type-I αβα 是 gapless source；ω=0 附近的大响应不是 clean insulating BPVE。
- Revised Figures 1 and 10 从 5 meV 起画；Figure 11 已删除全部 Type-I gapless row。
- Type-II 16.6 meV peak 不受这项排除影响。

Figure-specific convention

Figures 2-9: historical Plan-B renderings.

Signed response: current = - historical / 2.

Norm/envelope/absolute weight: current = historical / 2 (no sign).

Figures 1 and 10-12: already current-authority; do not convert again.

Current-authority anchors

- P2-approved η=8 component: +2.3678e4 μA·nm/V² at 16.6 meV。
- η=8 tensor-envelope norm: 4.7397e4 μA·nm/V²。
- η=1 largest scanned peak: 6.876e4 μA·nm/V² near 17.8 meV（diagnostic）。

Type-II 比 Type-I 明显更亮；错误零频不再展示

- 左上 x axis 从 5 meV 开始；αβα spurious zero-frequency maximum 被明确排除。
- 右上 αβγ 在 16.6 meV 达到 current envelope norm 4.7397e4 μA·nm/V²。
- Type-II active8/active12 overlap；Type-I remote active8 peak 不通过 window test。

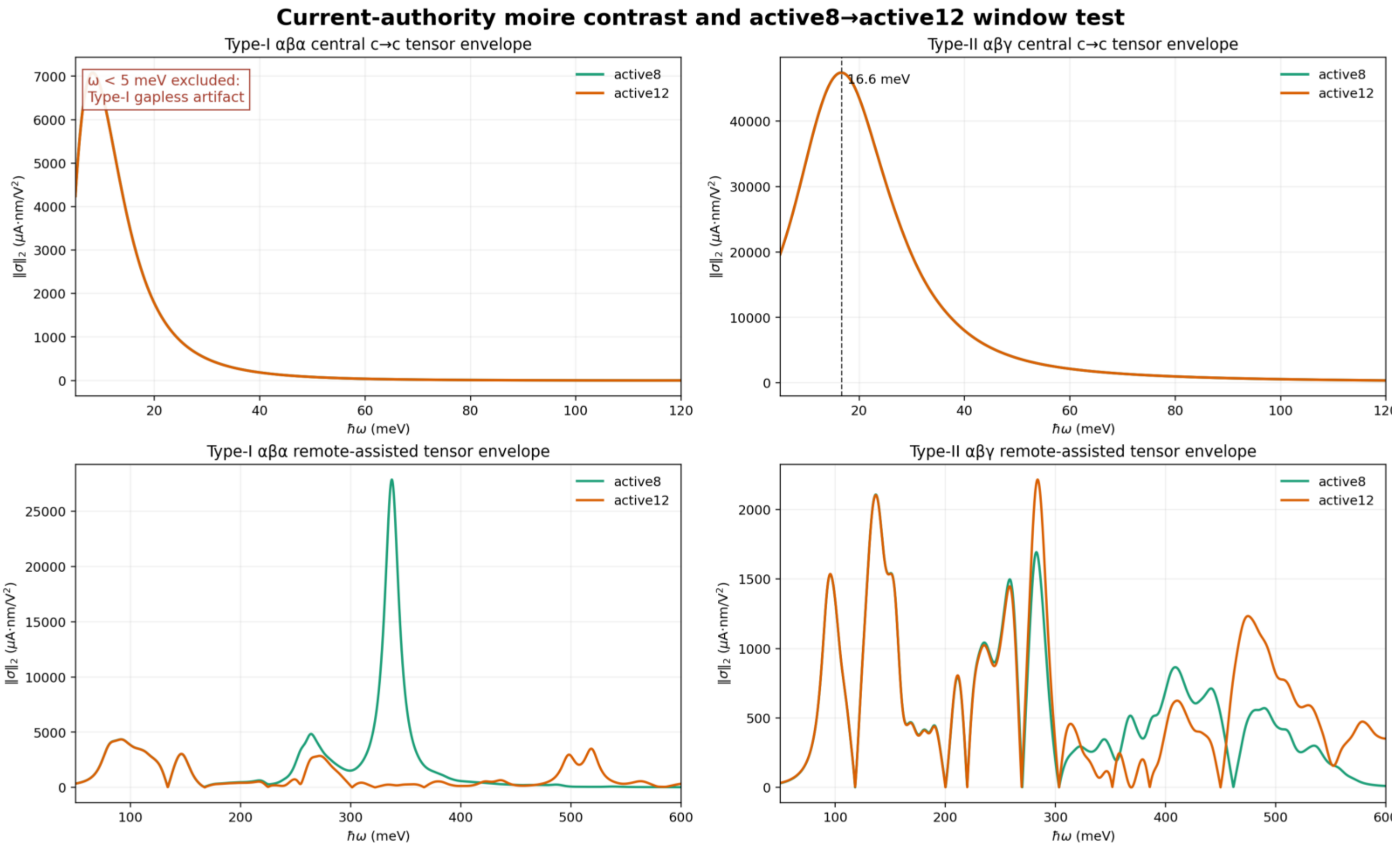


图 1 | Current-authority moire contrast；Type-I hbarω < 5 meV excluded。

最大扫描峰与 approved headline 必须分开

- Current magnitudes: η=1 → 6.876e4；η=2 → 4.97e4；η=3 → 4.19e4 μA·nm/V²。
- Largest scanned value 是 η=1 near 17.8 meV，但 peak height 强烈依赖 η。
- Approved headline 使用 η=8 P2 continuum component = 2.3678e4 μA·nm/V²。
- Direct/indirect gaps = 3.4795/1.5630 meV；candidate_near_degenerate_group=false。

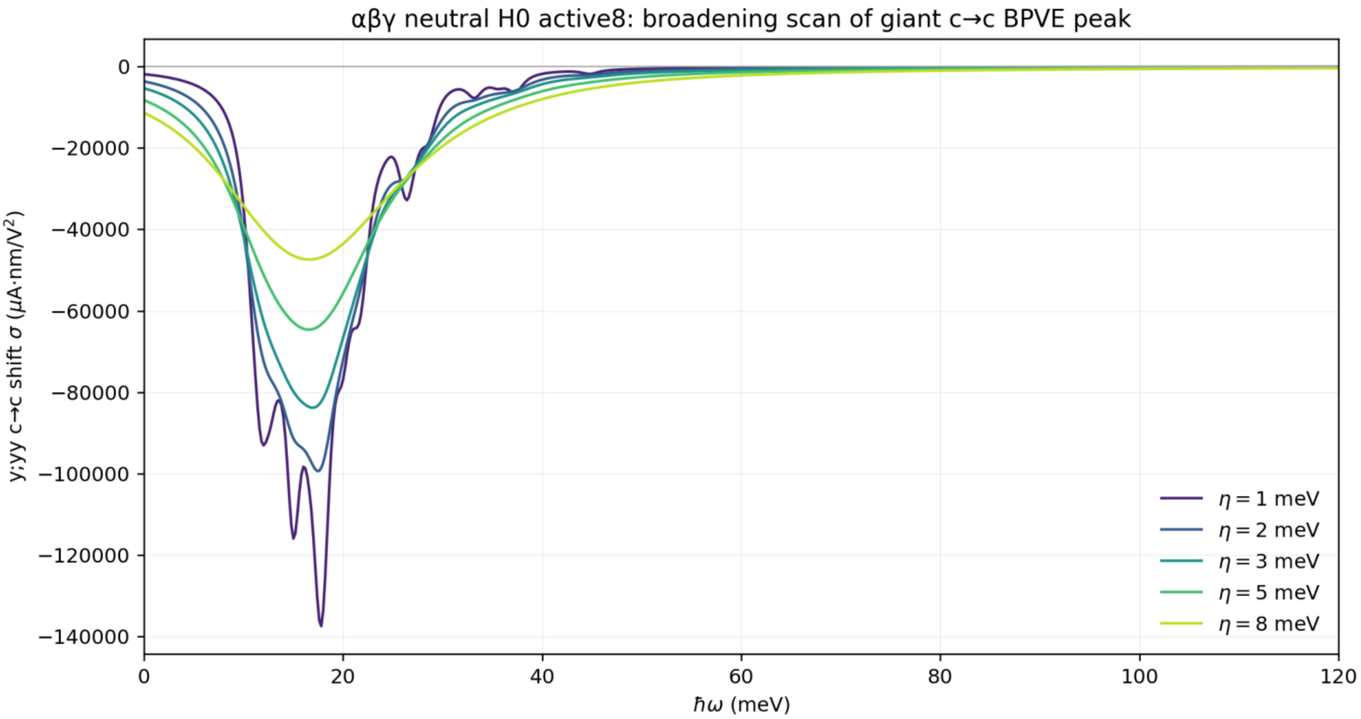


图 2 | Historical rendering of αβγ broadening scan；conversion shown in footer。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

18×18、24×24、30×30 曲线重合

- Peak position、line shape 与 integrated weight 在 mesh refinement 下稳定。
- 与 finite gaps 和 active8→active12 overlap 一起，排除离散网格偶然热点解释。

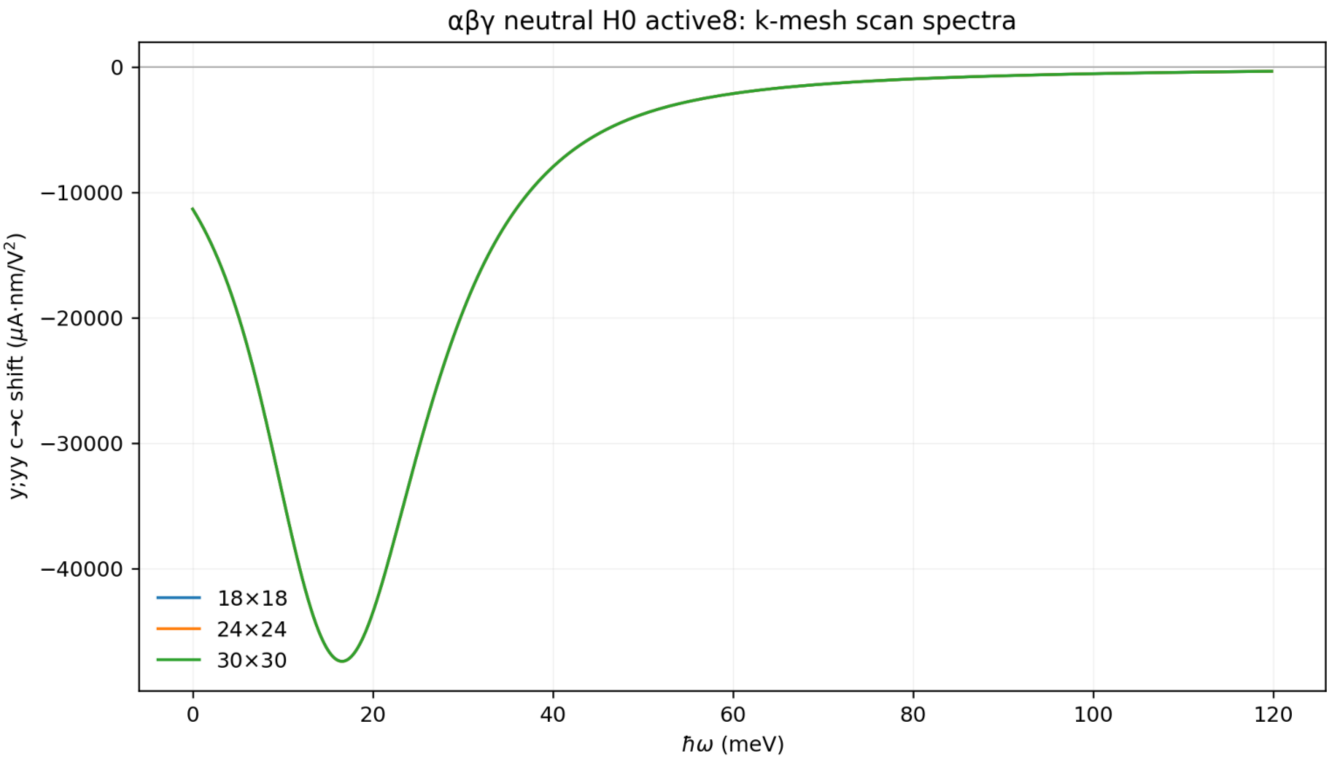


图 3 | αβγ η=8 k-mesh spectra。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

巨大响应是 signed tensor，不是单一正数

• 多个 symmetry-related components 同时很大并具有相反符号。

• Central c→c panel 重现低能 total：不是 remote-band artifact。

• 实验必须指定 polarization、current axis 与 structural domain。

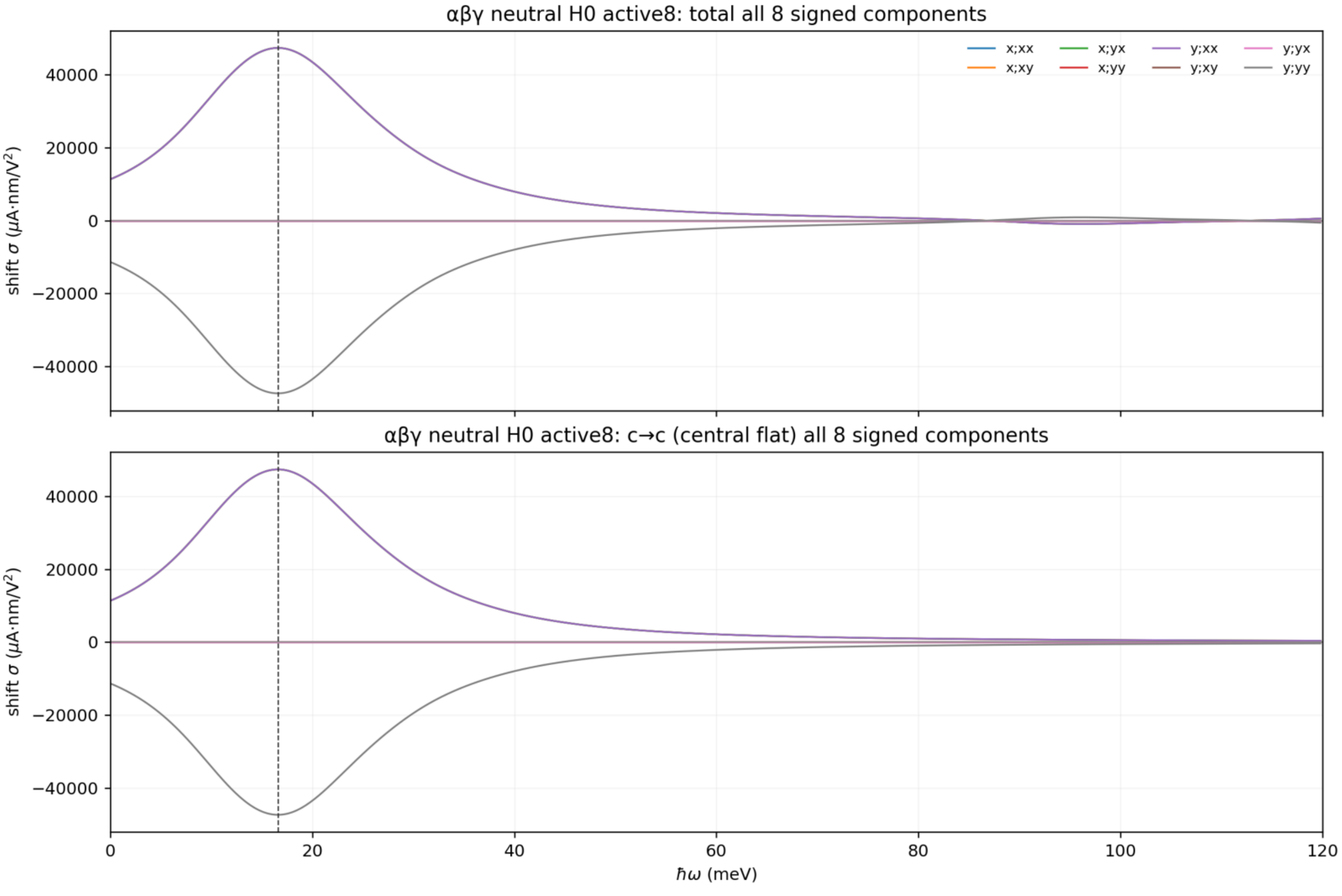


图 4 | Type-II αβγ 的八个 signed components。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

Mechanism：central-flat resonance，而非 remote artifact

- JDOS low-energy peak 与 dominant shift component 都由 central c→c channel 主导。
- Giant amplitude 还需要 optical weight、covariant shift/geometric factor 与有限 cancellation。

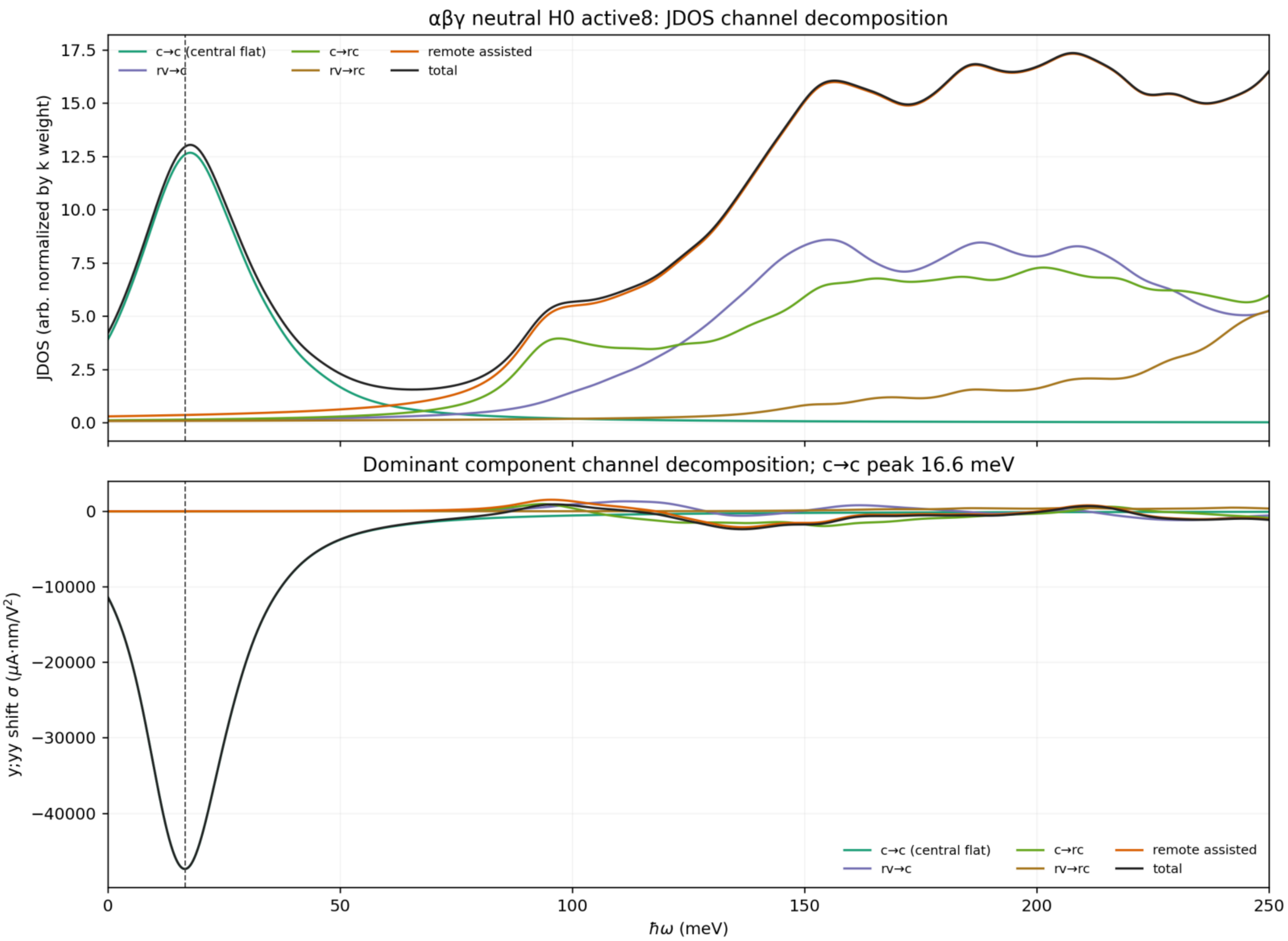
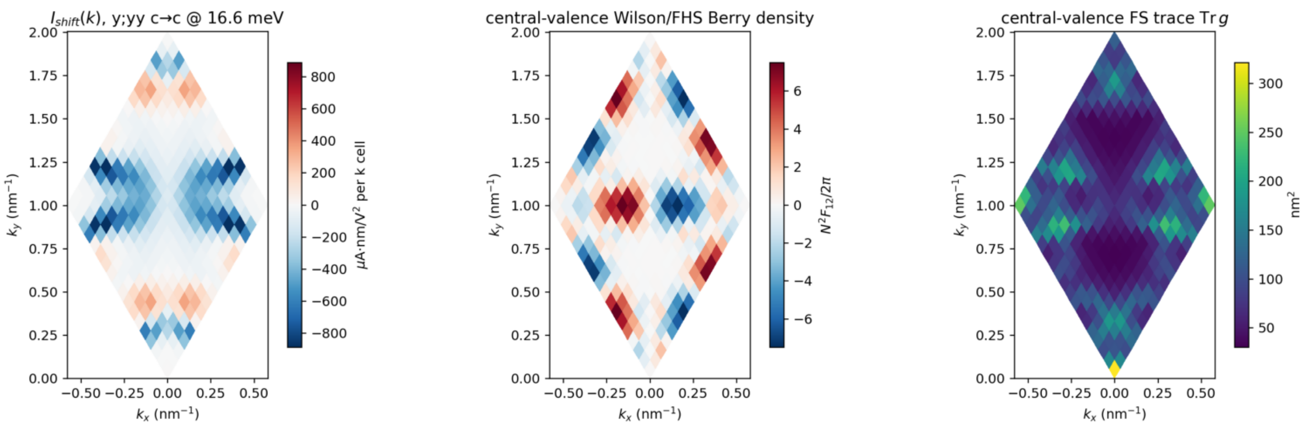


图 5 | JDOS and signed-channel decomposition。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

Local quantum geometry 是共同底层，不是 Chern 定标

- Berry/FS maps 是 occupied-bundle diagnostics，不是 pair-resolved transition geometry。
- FS trace 是 finite-difference subspace-fidelity diagnostic；co-location 不是 causal proof。
- 现有 evidence 只说明 Chern alone 不决定 BPVE magnitude。



Chern = (1/2π)∫BZ Ω(k)dk 是 unweighted global integral；

BPVE 是 frequency-resolved、transition-weighted、tensorial signed integral。

图 6 | Shift-current hotspot、Berry density 与 FS trace。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

不加相互作用的 peak 更大，但 HF 主要重排 spectral weight

- Neutral Type-II H0 η=8 peak 2.3678e4，是 ε=10 ν=0 HF peak 8.9e3 的约 2.66 倍。
- HF/H0 integrated low-energy-weight ratio = 0.905：相互作用主要 blue-shift/redistribute。
- Order 仍会改变 frequency、sign、polarization 与 c→c / c→rc channel content。

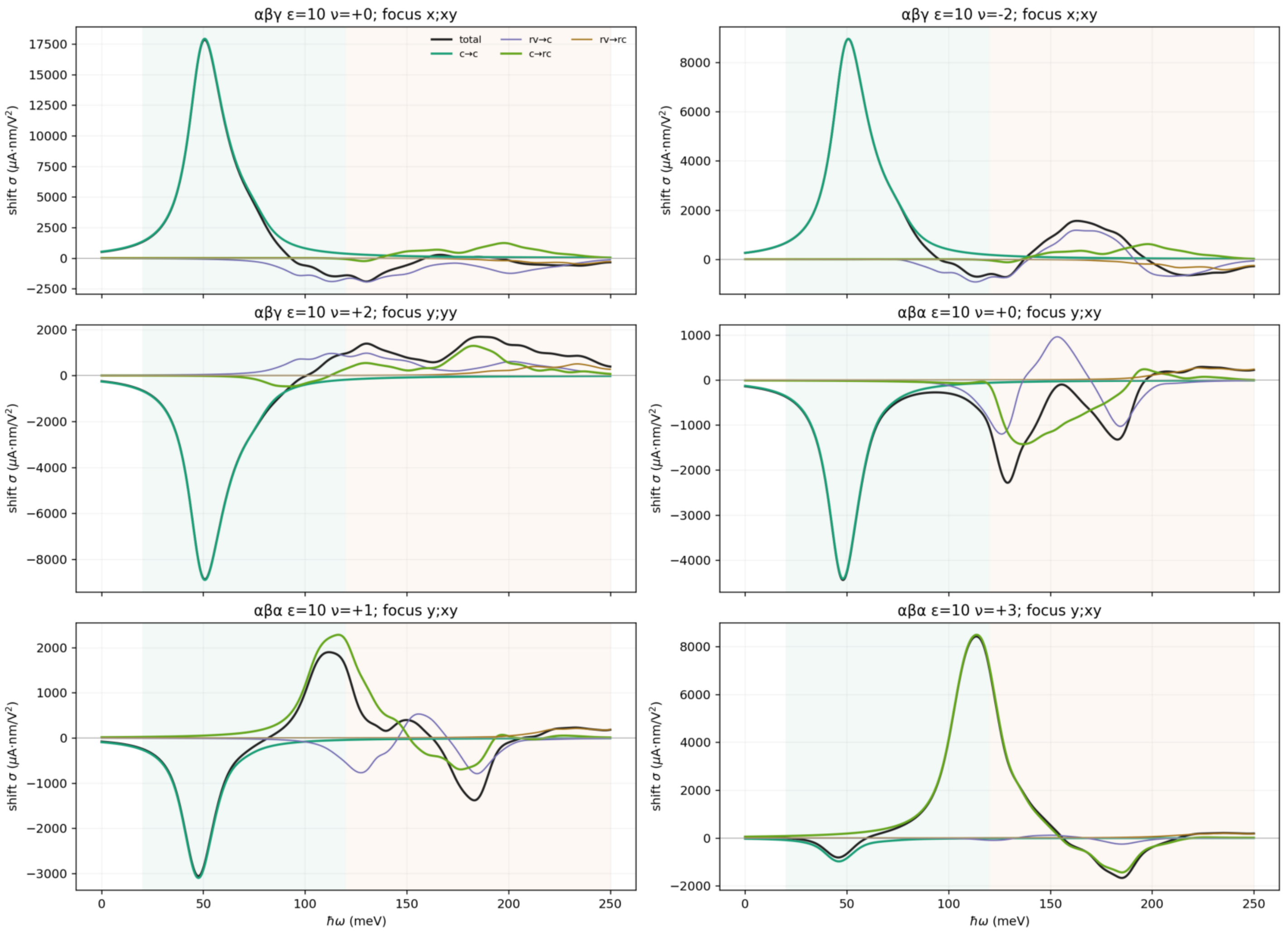


图 7 | active4 ε=10 representative HF states。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

Topology/order 相关，但 brightness 连续变化

- Type-II shown rows 有 active2/active4 Chern agreement：这是 active-space consistency。
- 它不是 thermodynamic stability；Type-I 集中了 Chern mismatches 与 IVC/KIVC order。
- 不同 Chern labels 不对应量子化 brightness steps。

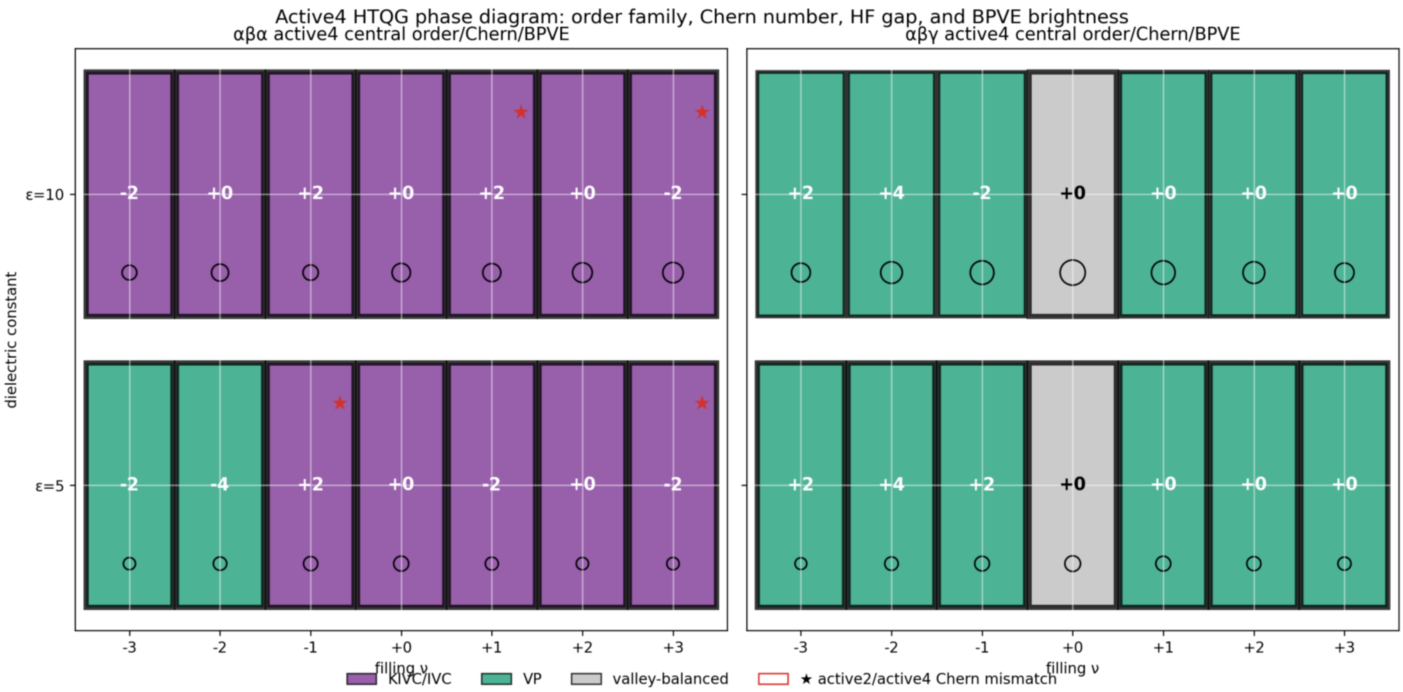


图 8 | Order family、Chern、gap 与 low-energy BPVE weight。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

减弱相互作用可增强 low-energy weight，但需保留 branch caveat

- Along sampled continuations, increasing dielectric ε generally increases Wlow and softens peaks。
- ε=8/12/15 seeded from ε=10；branch crossings 和 global ground state 未测试。
- 因此这是“弱相互作用有利”的支持 evidence，不是完整 phase-diagram theorem。

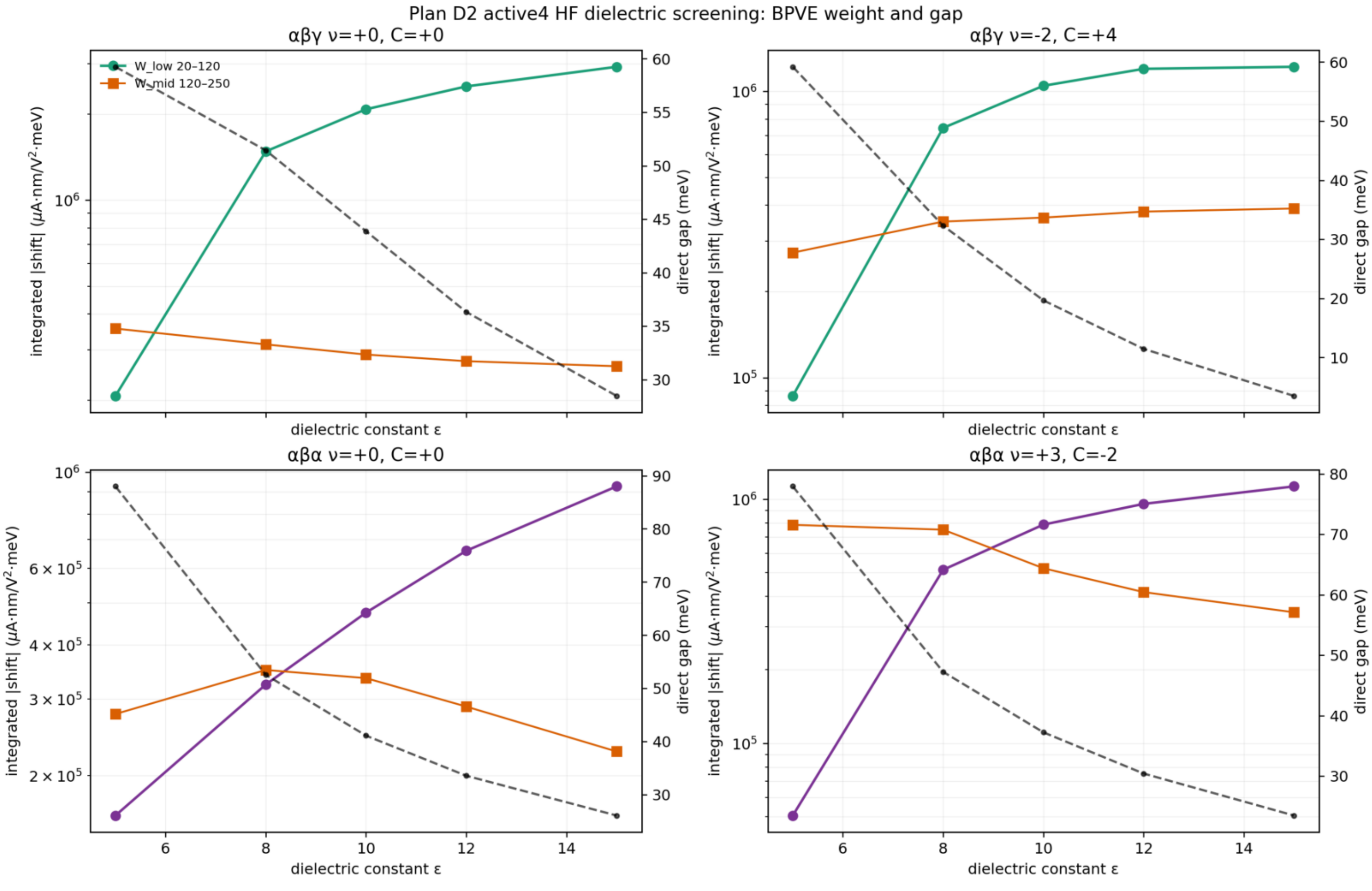


图 9 | Dielectric continuation scan。

Historical Figure 2-9 response scale: signed -> -old/2; norms/absolute weights -> old/2

单畴巨大，partner domains 等面积时几乎抵消

- Current-authority Type-II partner tensors 近乎 equal-and-opposite。
- Type-I ω<5 meV 已排除；剩余曲线仅作 gapless symmetry diagnostic。
- 增大 macroscopic current 的首要条件之一是 single-domain 或 strong domain imbalance。

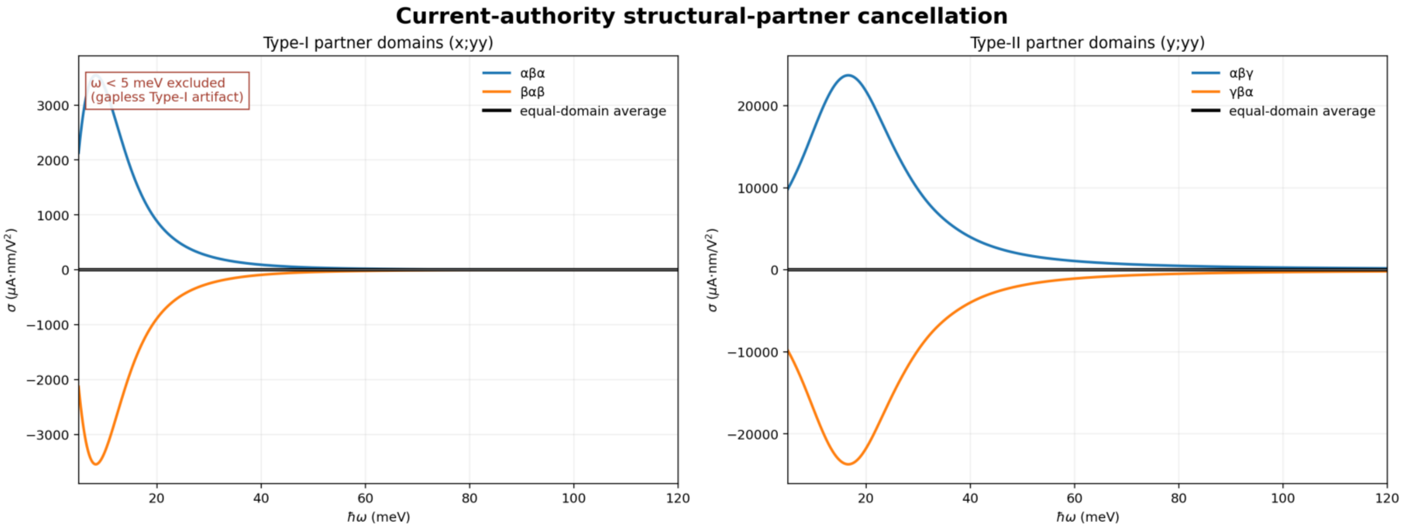


图 10 | Current-authority partner-domain cancellation；Type-I low-frequency exclusion。

只保留 clean Type-II polarization response

- Former Type-I gapless row 已删除，不再把错误 low-frequency coefficient 放入实验图。
- 选择 dominant polarization contraction 可最大化 single-domain current。
- Partner domain 反转 current vector；domain mapping 是实验判据。

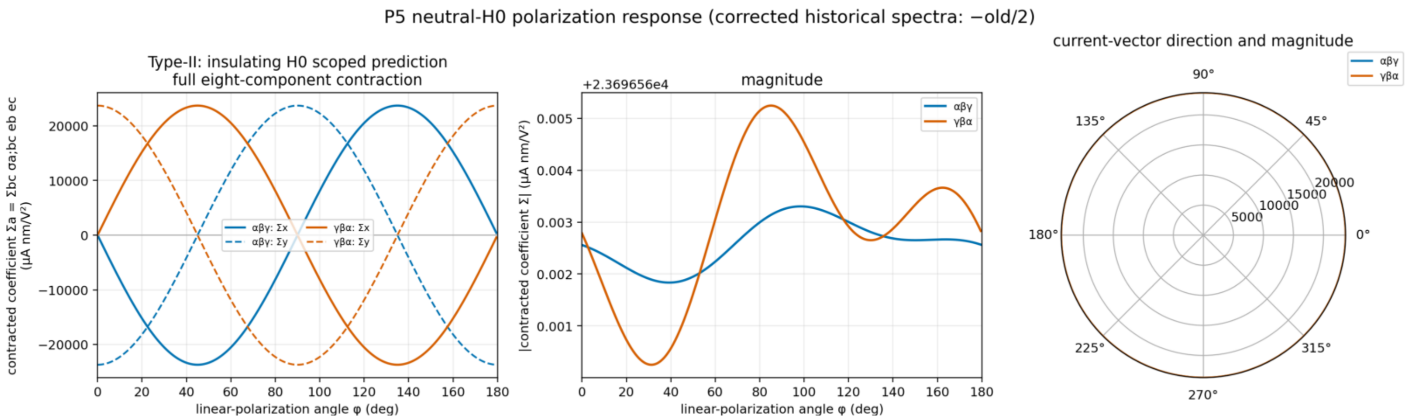


图 11 | Corrected Type-II full-tensor polarization contraction。

Largest peak、interaction effect、厚度换算与 ideal current

- Largest scanned η=1 peak = 6.876e4； P2-approved η=8 component = 2.3678e4 μA·nm/V²。
- H0 peak ≈ 2.66 × ε=10 ν=0 HF peak； smaller η sharpens peak but does not add oscillator strength。
- Effective bulk coefficient uses σ3D=σ2D/t_eff with t_eff=0.67-1.0 nm。

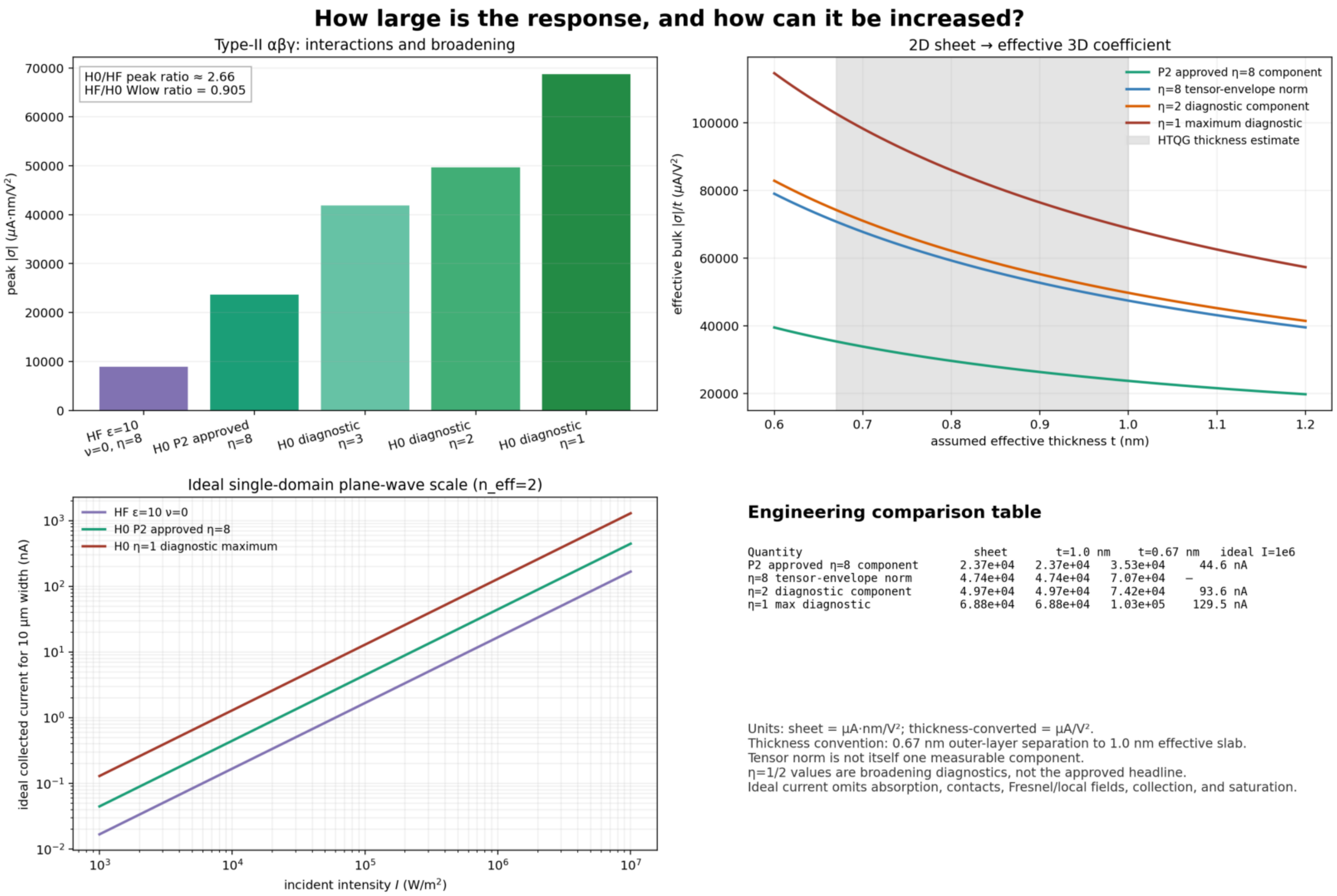


图 12 | All plotted quantities are current-authority; engineering conversions are explicitly assumption-bound。

如何让光电流变大，以及如何与 bulk materials 比较

Increase the intrinsic coefficient

- Choose Type-II αβγ near neutrality；avoid the invalid Type-I zero-frequency sector。
- Weaken interaction reshaping through screening；H0 peak is 2.66× the shown HF neutral peak。
- Use cleaner samples to reduce η, while reporting integrated weight alongside peak height。
- Align polarization and select one structural domain；equal partner domains cancel。

Increase collected current without changing σ

- Within perturbative response: increase incident intensity I and device width W。
- Ideal scale: j2D = σ2D × 10^-15 × I/(c ε0 n_eff), Icollected = j2D W。
- At I=1e6 W/m², n_eff=2, W=10 μm: approved → 44.6 nA；η=1 diagnostic → 129.5 nA。

Effective bulk coefficient

$$\sigma_{3D}^{eff} = \sigma_{2D}/t_{eff}$$

Three-layer convention: outer-layer center separation ≈ 0.67 nm；effective optical slab up to 1.0 nm。

P2-approved component: 2.37e4-3.53e4 μA/V²。

η=1 maximum diagnostic: 6.88e4-1.03e5 μA/V²。

Thickness conversion is conventional and inverse-thickness dependent.
It is for coefficient-level comparison, not a unique microscopic 3D property.

重写后的核心叙事与下一步

Validated story

- Structural selectivity：Type-II αβγ finite-frequency BPVE 明显强于 Type-I αβα。
- Legitimacy：finite gaps/frequency + broadening + mesh + active-window checks。
- Mechanism：central c→c resonance × optical/geometric factor × signed cancellation。
- Interaction：H0 peak 更高；HF shifts/redistributes weight；screening can recover low-energy brightness。
- Experiment：single-domain、polarization-resolved THz current；避免 Type-I ω≈0 artifact。

Needed causal checks

- Matched Type-I/II factorization: JDOS, $|r|^2$, covariant shift term, cancellation。
- Same-order/different-C and same-C/different-order HF comparisons。
- Device-specific absorption, local field, contacts, collection, relaxation, saturation。
- Literature comparison using the same σ definition, broadening, thickness, and polarization convention。

Active6 Type-I P1-P5 remains a controlled supplement, not the giant-BPVE headline.

边界、provenance 与 references

- Type-I hbarω<5 meV is excluded because the neutral H0 source is gapless and the zero-frequency response is invalid。
- η=1/2/3 maxima are broadening diagnostics；approved headline is η=8 P2 continuum component。
- Tensor-envelope norm is not itself one measurable tensor component。
- 0.67-1.0 nm effective thickness gives a comparison convention, not a unique 3D material property。
- Ideal current omits Fresnel/local fields, absorption, contacts, collection, relaxation, and saturation。
- Only αβγ is directly P2 full-continuum approved；γβα is projected-active8 partner。

Figure provenance

Figures 2-9: exact historical Plan-B products.
Figures 1 and 10: current-authority re-plots of saved spectra with Type-I <5 meV excluded.
Figure 11: corrected P5 Type-II crop. Figure 12: formula-derived engineering summary.

Selected references

Sipe & Shkrebtii, Phys. Rev. B 61, 5337 (2000).

Morimoto & Nagaosa, Sci. Adv. 2, e1501524 (2016).

Chaudhary et al., Phys. Rev. Research 4, 013164 (2022).

Package: narrative + 12 PNG figures only. No NPZ/JSON/CSV/checkpoints/HF density.